

Root Grip Erosion Rescue

Can roots hold a slope together during a storm?

Win condition: Most soil retained after standard pour.

THE SCIENCE YOU ARE USING

Roots anchor soil. Bare soil on a slope washes away fast when rain hits it. Roots grip the soil and hold it in place, and a denser, deeper root network holds better. This is why planting vegetation is a real tool for stopping erosion.

Slope and runoff. Steeper slopes and bare surfaces shed water and soil quickly. You build a mini slope with root-like anchors, pour a standard amount of water, and measure how much soil stays put.

HOW TO BUILD AND RUN IT

- 01 Build a bare slope.** Pack soil into a tray at a set angle with no roots. This is your control.
- 02 Add root anchors.** Push string or yarn roots into a second slope to mimic a planted hillside.
- 03 Pour the standard storm.** Pour the same measured water on each slope from the same height.
- 04 Measure soil retained.** Compare how much soil washed out of each tray. Record the difference.
- 05 Redesign the root network.** Add more or deeper roots and re-test to retain more soil.

Safety: Work over a tray and wear gloves to handle wet soil and muddy water; wipe spills right away so the floor stays dry. Pour muddy water into the slop bucket, not the sink. No goggles are needed for pouring water down a soil tray. Allergy: Dust or pollen sensitivity, so the soil is sieved and dampened to keep dust down.

MATERIALS

Trays	2 per team (8)
Soil	shared
String or yarn roots	shared
Craft sticks	shared
Watering cups	1 per team (4)

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE thing we changed for the redesign (more roots, deeper roots, or a denser pattern): _____

Soil retained, baseline (bare vs rooted slope): _____

Soil retained, after redesign: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Soil retained	40
Runoff control	25
Design explanation	20
Cleanup	15
Total	100